

Central Line Dressing Change Policy

CLABSI Prevention: Sterile Technique, Frequency, Assessment & Documentation

Regulation: 42 CFR §482.42 | CDC CLABSI Prevention Guidelines | INS Standards

PURPOSE & SCOPE

Central Line-Associated Bloodstream Infections (CLABSIs) are largely preventable and are a significant cause of patient morbidity, mortality, and cost. This policy establishes the standardized procedure for central venous catheter (CVC) dressing changes, including PICCs, tunneled catheters, implanted ports, hemodialysis catheters, and temporary CVCs. It applies to all licensed clinical staff performing central line care across all clinical settings.

Zero Tolerance: CLABSI Prevention is a Patient Safety Priority

CLABSIs are a CMS Hospital-Acquired Condition (HAC) that is non-reimbursable. CLABSI rates are publicly reported on Hospital Compare.

Every CLABSI must be reviewed by the infection control and quality teams. Root cause analysis is required for all CLABSIs.

Strict adherence to this policy is mandatory. Shortcuts to sterile technique are never acceptable, regardless of time constraints or circumstances.

CATHETER TYPES & DRESSING REQUIREMENTS OVERVIEW

Catheter Type	Dressing Type	Change Frequency	Technique
Non-tunneled CVC (internal jugular, subclavian, femoral)	Transparent semipermeable membrane (TSM)	Every 7 days or when soiled, loose, or non-intact	Sterile — full sterile technique
PICC (Peripherally Inserted Central Catheter)	TSM (preferred) or gauze	TSM: every 7 days; Gauze: every 48 hours or when soiled	Sterile — full sterile technique
Tunneled CVC (Hickman, Broviac)	TSM after healed exit site	Every 7 days or per manufacturer; may be uncovered once healed per physician order	Sterile technique
Implanted Port (accessed)	TSM over non-coring needle	Every 7 days while accessed; change needle every 7 days	Sterile — port access requires sterile technique
Hemodialysis CVC	TSM or gauze per dialysis protocol	Every dialysis session (dialysis staff) or every 48–72 hrs if between sessions	Sterile — mask required for patient and nurse

Arterial Line (radial/femoral)	TSM	Every 7 days or when soiled/loose	Sterile technique
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EQUIPMENT & SUPPLY PREPARATION

Use a standardized central line dressing change kit whenever available. Verify all items before opening. Inspect for integrity, sterility indicator, and expiration date.

Required Supply	Notes
Sterile gloves (correct size)	Non-sterile (clean) gloves for initial dressing removal
Central line dressing change kit or:	Sterile drape / sterile field
Chlorhexidine gluconate (CHG) 2% applicator	(preferred) or 70% isopropyl alcohol if CHG allergy/contraindication
Chlorhexidine-impregnated sponge disk (Biopatch)	For use at catheter-skin insertion site — evidence-based CLABSI reduction
Transparent semipermeable membrane dressing (TSM)	e.g., Tegaderm CHG or standard Tegaderm per protocol
Sterile gauze 2×2 squares (2–3)	For cleaning and to cover site if CHG gel present
Engineered stabilization device (StatLock, SecureAcath)	Or sterile tape strips if sutures absent/removed
Mask — surgical mask for nurse	Mask for patient (patient must wear mask and turn head away during procedure)
Label and indelible marker	For dressing date/time labeling
Sharps container accessible	If removing sutures

STEP-BY-STEP DRESSING CHANGE PROCEDURE

This procedure requires strict sterile technique throughout. Any break in sterile technique requires restarting with new supplies.

1	Verify Order and Assess Need Confirm a physician order for central line dressing change (or that the change is per standing protocol). Assess dressing: is it due by schedule, or is it soiled, loose, lifting, bloody, or non-occlusive? Any of these conditions require immediate dressing change regardless of schedule.
2	Perform Hand Hygiene Perform hand hygiene with ABHR or soap and water per WHO Five Moments protocol. This is Moment 2 (before an aseptic procedure). Do NOT skip this step even if you will be wearing gloves.

3	Gather Supplies and Prepare Environment Gather all required supplies before entering the room. Raise the bed to working height. Position patient for access to catheter insertion site. Ensure adequate lighting. Close room door or curtain. Turn patient's face away from the insertion site.
4	Apply Masks Apply a surgical mask over your nose and mouth. Instruct the patient to apply a surgical mask (or turn their face away and avoid talking, coughing, or sneezing toward the insertion site during the procedure). This is a required CLABSI prevention measure.
5	Open Sterile Supplies Using Aseptic Technique Open the dressing change kit and all supplies onto a clean, dry surface using aseptic non-touch technique. Do NOT contaminate the inner sterile field. Open CHG applicator, Biopatch, and TSM dressing packaging and place on the sterile field without touching sterile surfaces with ungloved hands.
6	Don Clean Gloves and Remove Old Dressing Apply clean (non-sterile) gloves. Remove the old dressing by peeling edges toward the insertion site (to avoid dislodging the catheter). Stabilize the catheter hub with your non-dominant hand while removing dressing. Note: dressing removal direction matters — always toward the catheter, never away. Assess old dressing for: drainage, odor, blood, condition of Biopatch.
7	Remove Clean Gloves and Perform Hand Hygiene Remove and discard clean gloves. Perform hand hygiene with ABHR immediately before donning sterile gloves. This is a critical step that is frequently skipped — it is mandatory.
8	Assess the Insertion Site Without touching the site, visually inspect and palpate (through intact skin) for: erythema, swelling, warmth, induration, tenderness, purulent drainage, or tunnel tract tenderness. Document findings. If signs of infection are present, complete the dressing change, notify the physician, and document. Do NOT remove the catheter without a physician order.
9	Apply Sterile Gloves Don sterile gloves using open or closed glove technique, maintaining sterility throughout. Verify sterile glove integrity before proceeding. If sterility is compromised, obtain new sterile gloves.
10	Skin Antisepsis with Chlorhexidine Gluconate (CHG) Open the CHG 2% applicator (Chloraprep). Apply to insertion site and surrounding skin using a back-and-forth scrubbing motion for a minimum of 30 seconds. Apply to an area of at least 3–4 cm radius around the insertion site. Allow to air dry completely — MINIMUM 30 seconds for CHG; do NOT fan, blow, or blot dry. CHG MUST be completely dry before applying the dressing; wet CHG can cause skin irritation and reduces efficacy. CHG Allergy/Contraindication: Use 70% isopropyl alcohol scrub × 30 seconds and allow to dry completely.

11	Apply Chlorhexidine-Impregnated Sponge Disk (Biopatch) Place the Biopatch disk with the CHG-impregnated (blue/darker) side DOWN, directly against the skin at the catheter insertion site. The disk should lie flat against the skin. Slot the disk around the catheter — do not force. Replace Biopatch with every dressing change. Contraindications: known CHG allergy, neonates under 7 days, premature infants (skin sensitivity risk).
12	Apply Engineered Stabilization Device If using a StatLock or similar engineered stabilization device (ESD), apply now per manufacturer instructions. ESDs are preferred over sutures for catheter stabilization — they reduce catheter movement (which increases infection and thrombosis risk) without skin puncture. Replace ESD with every dressing change.
13	Apply Transparent Semipermeable Membrane (TSM) Dressing Apply the TSM dressing (e.g., Tegaderm CHG, Tegaderm) over the Biopatch and insertion site. Center the dressing over the insertion site. Apply without tension or stretch (to prevent MARS — Medical Adhesive-Related Skin Injury). Press firmly from the center outward to eliminate air bubbles and ensure complete adhesion. The dressing must be fully occlusive on all edges with no lifting.
14	Secure Catheter Tubing Loop excess catheter tubing and secure to the dressing with sterile tape. Ensure catheter hub is accessible for connection/disconnection. Avoid kinking. Label tubing with catheter type, lumen designation, and change date per policy.
15	Label the Dressing Label the dressing with: date of dressing change, time, nurse initials. Place label on the edge of the dressing — never over the site, as this impairs visualization.
16	Remove Gloves and Perform Hand Hygiene Remove sterile gloves and perform hand hygiene with ABHR immediately after the procedure (WHO Moment 3).
17	Dispose of Supplies and Return Environment Dispose of all used supplies per biohazardous waste policy. Remove mask. Return bed to safe position. Ensure call light is within reach. Perform hand hygiene.
18	Document in the Medical Record Documentation must include: date/time of dressing change, condition of old dressing, insertion site assessment findings, antiseptic agent used, dressing type applied, Biopatch applied (yes/no), stabilization device used, any concerns or complications, and patient tolerance.

INSERTION SITE ASSESSMENT — GRADING & ESCALATION

Assessment Grade	Clinical Findings	Required Action
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Grade 0 — No signs	No redness, swelling, tenderness, or drainage	Routine documentation; continue standard dressing changes
Grade 1 — Phlebitis/early infection	Mild erythema ≤2 cm, mild tenderness at insertion site, no drainage	Increase monitoring frequency; notify charge nurse; document trend; physician notification if worsening
Grade 2 — Suspected infection	Erythema >2 cm, swelling, warmth, tenderness, purulent or cloudy drainage	Notify physician immediately; obtain cultures (blood ×2 sets, site swab if drainage present); do NOT remove line without order; document thoroughly
Grade 3 — Confirmed/severe infection	Purulent drainage, fluctuance, extensive erythema, systemic signs (fever, hypotension)	STAT physician notification; prepare for line removal per order; blood cultures ×2; initiate sepsis protocol if systemic signs present
Tunnel/Port Infection	Erythema, tenderness, swelling along catheter tunnel or over port pocket	Notify physician; likely requires catheter removal — tunneled catheter/port infections typically cannot be treated without removal

CLABSI PREVENTION BUNDLE — REQUIRED DAILY CHECKLIST

The CLABSI prevention bundle must be assessed daily for all patients with central venous catheters. Bundle compliance must be documented in the nursing assessment.

Bundle Element	Requirement
Daily necessity review	Is the central line still needed? Document justification for continued use. Prompt physician to remove line if no longer clinically necessary.
Hand hygiene before all line access	Hand hygiene must be performed before every central line access — flushes, blood draws, tubing changes, cap changes.
Scrub the hub	All catheter hubs and needleless connectors must be scrubbed with 70% isopropyl alcohol × 15 seconds and allowed to dry × 15 seconds before any access.
Dressing integrity check	Assess dressing every shift: is it clean, dry, occlusive, and labeled? Change if loose, soiled, bloody, or due by schedule.
Tubing change compliance	IV administration sets must be changed per policy (standard: every 96 hours; lipid-containing solutions: every 24 hours; blood products: after each unit or per 4 hours).
Needleless connector (NLC) change	NLCs must be changed per policy (typically every 96 hours or with each tubing change). Cap replacement requires ABHR first.
CHG bathing	Patients with central lines must receive daily CHG (chlorhexidine gluconate) 2% bathing cloths to reduce skin flora colonization. Document completion daily.
Head of bed elevation	Unless contraindicated, maintain HOB at 30–45 degrees for patients with CVCs to reduce aspiration and VAP risk (if also ventilated).

DOCUMENTATION REQUIREMENTS

The following elements must be documented in the patient's electronic health record at the time of dressing change and with each shift assessment:

- Catheter type, size, and insertion date
- Insertion site location (right/left IJ, subclavian, femoral, PICC arm/side, port)
- Date and time of dressing change
- Condition of previous dressing (intact, soiled, bloody, loose)
- Insertion site assessment finding and grade (0–3)
- Antiseptic agent used (CHG 2% or alcohol if CHG contraindicated — reason documented)
- Biopatch applied: Yes / No (if No, document reason)
- Dressing type applied (TSM type, gauze if applicable)
- Stabilization device type and condition
- Patient tolerance and any complications
- CLABSI bundle compliance (daily checklist completed)
- Daily line necessity reviewed and documented
- Nurse signature and credentials

Line Days Tracking — Required for CLABSI Surveillance

Central line days must be tracked daily as part of NHSN CLABSI surveillance. Report monthly central line days and any CLABSI events to the Infection Preventionist.

CLABSI rate = (Number of CLABSIs ÷ Central Line Days) × 1,000. Target: Standardized Infection Ratio (SIR) ≤1.0. Report to QAPI monthly.